

RFC OTA-Based LDO Voltage Regulator — UMC 180 nm CMOS

M.Tech Thesis Project | NIT Silchar | Karnati Nagaraju

A comparative design and simulation study of two Low-Dropout (LDO) voltage regulators using the Recycling Folded Cascode (RFC) OTA as the error amplifier, implemented in UMC 180 nm CMOS technology using Cadence Virtuoso/Spectre.

Overview

LDO regulators for battery-powered IoT sensor nodes face a fundamental three-way conflict: output voltage precision, loop stability, and quiescent current consumption — all coupled through the error amplifier transconductance and bias current.

This work proposes a **High-Swing RFC OTA with self-biased cascode architecture (Design 2)** and benchmarks it against a conventional RFC OTA baseline (Design 1). The key innovation is a self-biased cascode current mirror that simultaneously improves output voltage swing (~120 mV recovery), provides automatic PVT compensation, and eliminates parasitic bias current overhead — without external trimming.

Specifications

Parameter	Value
Technology	UMC 180 nm CMOS (1P6M)
Supply voltage (V _{DD})	1.8 V
Regulated output (V _{out})	1.2 V
Max load current	50 mA
Load capacitance	1 μ F
Feedback resistors	R1 = 50 k Ω , R2 = 100 k Ω
Pass transistor	W/L = 500 μ m / 1 μ m (PMOS)
Reference voltage	0.6 V (ideal)
Simulator	Cadence Spectre (BSIM3)

Design Comparison

Parameter	Design 1 (Baseline RFC)	Design 2 (HS-RFC)	Change
Phase margin (TT, nominal)	61.49°	87.95°	+43%
DC open-loop gain	71.95 dB	68.45 dB	−3.5 dB
GBW	7.61 MHz	2.51 MHz	−3×
Load regulation	19.75 μV/mA	7.98 μV/mA	−60%
PSRR @ 1 kHz	−41.57 dB	−46.62 dB	+5.1 dB
Quiescent current	66.32 μA	72.46 μA	+9.2%
Settling time	0.73 μs	8.06 μs	+11×
Transient undershoot	29 mV	67 mV	—
Current efficiency	99.87%	99.86%	≈same
Transistor sizing	Strong inversion	Moderate inversion (gm/ID = 17–19 V ^{−1})	—
Biasing	Static external	Self-biased cascode	—

Design 2 achieves the **highest phase margin and lowest load regulation** among 9 published LDO designs compared in the thesis (Table 5.8).

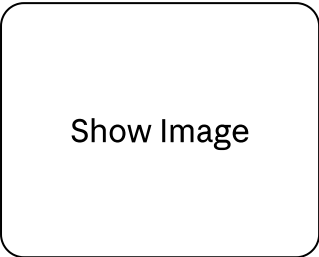
PVT Robustness (Design 2)

Full 27-point factorial analysis: 3 process corners (SS, TT, FF) × 3 supply voltages (1.62, 1.80, 1.98 V) × 3 temperatures (−40°C, 27°C, 125°C).

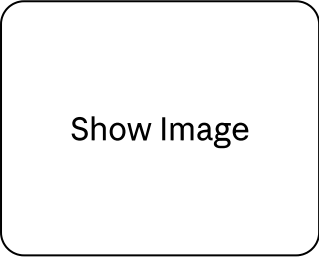
Metric	Result
Phase margin variation (all 27 PVT)	0.50°
Minimum phase margin (worst-case)	86.52° @ SS, 1.62 V, −40°C
Worst-case load regulation	48.42 μV/mA @ 125°C
Best-case PSRR @ 1 kHz	−48.71 dB

Schematics

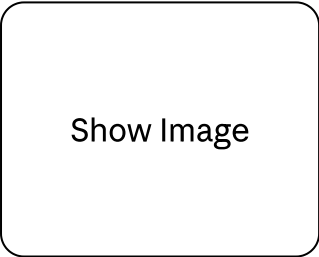
LDO System Architecture



Design 1 — Baseline RFC OTA (Static External Biasing)

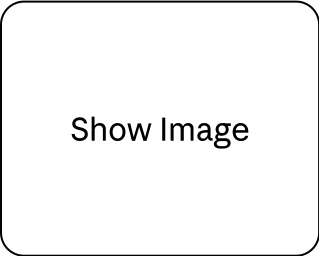


Design 2 — Proposed High-Swing RFC OTA (Self-Biased Cascode)

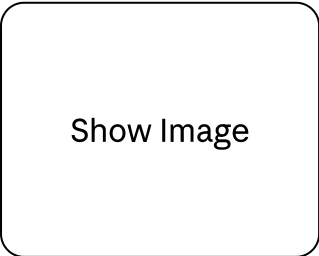


Simulation Results

Frequency Response — Design 1 vs Design 2



PSRR — Design 1 vs Design 2



Transient Response — Load Step 1 mA → 50 mA

Show Image

Stability (Full Load) — Process Corners

Show Image

Stability (Full Load) — Temperature Variation

Show Image

PSRR — Process Corners

Show Image

PSRR — Temperature Variation

Show Image

Key Technical Contributions

1. **Novel self-biased high-swing cascode biasing for RFC OTA-based LDOs** — cascode gate voltages self-generated from the circuit's own bias network using diode-connected devices.

Recovers ~120 mV output swing, provides automatic PVT tracking, eliminates parasitic bias current from external resistors.

2. **First documented gm/ID sizing procedure for RFC OTA in LDO context** — systematic step-by-step procedure sizing RFC OTA transistors in moderate inversion using PDK lookup tables (UMC 180 nm). Achieves 40–50% reduction in drain current at the same transconductance target vs. strong inversion.
3. **Comprehensive 27-point PVT validation** — full factorial PVT analysis demonstrating 0.50° total phase margin variation, considerably more thorough than typical 3-corner reports in published literature.
4. **Quantitative speed-stability trade-off characterisation** — two RFC OTA variants occupying different regions of the speed-stability-power design space compared in a structured manner.

Publication

Karnati Nagaraju, Sourav Chanda, and Dr. Bijit Choudhuri, "Design of a High-Precision, High-Stability Low-Dropout Regulator Using High-Swing Recycling Folded Cascode OTA," in *Proc. International Conference on Emerging Trends in Technology, Engineering, Computing and Networking (E2TECN 2026)*, SVNIT Surat, April 2026. **Published in Springer Lecture Notes in Electrical Engineering (LNEE). [Presented]**

Limitations

- **Simulation only** — all results from schematic-level Cadence Spectre simulation. No physical layout, post-layout extraction, or silicon fabrication.
 - **Ideal voltage reference** — V_{ref} implemented as a fixed 0.6 V ideal source. A real bandgap reference would add temperature drift and noise.
 - **Proprietary PDK** — UMC 180 nm CMOS PDK (BSIM3 models) is not distributable. Netlists are not included.
 - **No startup circuit** — design assumes bias currents are correctly established at power-on.
 - **Model-to-hardware correlation** — errors of 5–10% are typical for circuits of this complexity.
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Repository Structure

```
RFC-OTA-LDO-UMC180nm/
```

```
|— README.md
```

```
|— docs/
```

```
|   |— thesis.pdf
```

```
# M.Tech thesis (NIT Silchar, 2025-26)
```

```
|   └─ E2TECN_Manuscript_Nagaraju.pdf  # Conference paper (Springer LNEE)
└─ schematics/
    │   └─ LDO_SCHEMATIC_FINAL1.jpg      # System-level LDO architecture
    │   └─ BASELINE_OTA_final.jpg        # Design 1: Conventional RFC OTA
    │   └─ PROPOSED_OTA_FINAL.jpg        # Design 2: High-Swing RFC OTA
└─ results/
    │   └─ gain_GBW_PM_of_design1__2.jpeg
    │   └─ psrr_of_both_designs_.png
    │   └─ transient_final_v7_1_.png
    │   └─ phase_full_process.png
    │   └─ phase_full_temp.png
    │   └─ phase_light_process.png
    │   └─ phase_light_temp.png
    │   └─ phasemargin_full_voltage.png
    │   └─ phasemargin_light_voltage.png
    │   └─ psrr_process.png
    │   └─ psrr_temp.png
    │   └─ psrr_voltage.png
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Author

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